Energy, Work & Power

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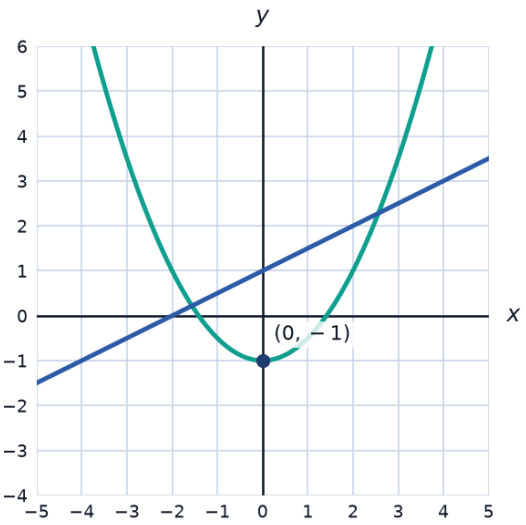
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Work Done by Constant & Inclined Forces

A constant force of 15 N is applied to an object, causing it to move a distance of 4 m in the direction of the force. Calculate the work done by the force.

2 [Foundational] • Target: Kinetic Energy & Potential Energy

A force of 42 N acts at 30° to the direction of motion of a body that moves 27 m. Find the work done, leaving your answer in exact (surd) form.



3 [Foundational] • Target: The Work-Energy Principle

A force of 20 N acts at 45° to the direction of motion of a body that moves 10 m. Find the work done, leaving your answer in exact (surd) form.

4 [Foundational] • Target: Power and Vehicles in Motion

A force of 28 N acts at 45° to the direction of motion of a body that moves 28 m. Find the work done, leaving your answer in exact (surd) form.

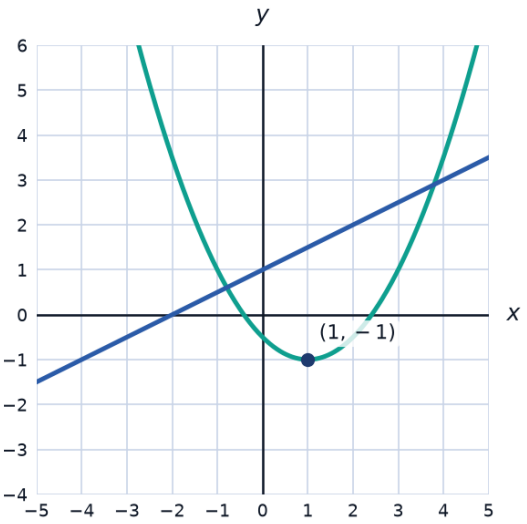
5 [Foundational] • Target: Work Done by Constant & Inclined Forces

A force of 42 N acts at 45° to the direction of motion of a body that moves 14 m. Find the work done, leaving your answer in exact (surd) form.

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6 [Foundational] • Target: Kinetic Energy & Potential Energy

A force of 40 N acts at 30° to the direction of motion of a body that moves 15 m. Find the work done, leaving your answer in exact (surd) form.



7 [Foundational] • Target: The Work-Energy Principle

A force of 75 N acts at 45° to the direction of motion of a body that moves 8 m. Find the work done, leaving your answer in exact (surd) form.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Power and Vehicles in Motion

A force of 78 N acts at 45° to the direction of motion of a body that moves 4 m. Find the work done, leaving your answer in exact (surd) form.

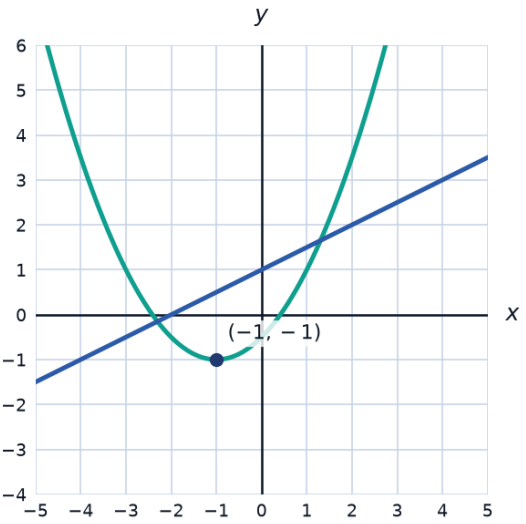
9 [Intermediate] • Target: Work Done by Constant & Inclined Forces

A force of 64 N acts at 60° to the direction of motion of a body that moves 19 m. Find the work done, leaving your answer in exact (surd) form.

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10 [Intermediate] • Target: Kinetic Energy & Potential Energy

A force of 57 N acts at 60° to the direction of motion of a body that moves 21 m. Find the work done, leaving your answer in exact (surd) form.



11 [Intermediate] • Target: The Work-Energy Principle

A force of 38 N acts at 60° to the direction of motion of a body that moves 23 m. Find the work done, leaving your answer in exact (surd) form.

12 [Intermediate] • Target: Power and Vehicles in Motion

A force of 77 N acts at 60° to the direction of motion of a body that moves 7 m. Find the work done, leaving your answer in exact (surd) form.

13 [Intermediate] • Target: Work Done by Constant & Inclined Forces

A body of mass 11 kg moving at 7 m/s has work 356 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

14 [Intermediate] • Target: Kinetic Energy & Potential Energy

A body of mass 11 kg moving at 1 m/s has work 789 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

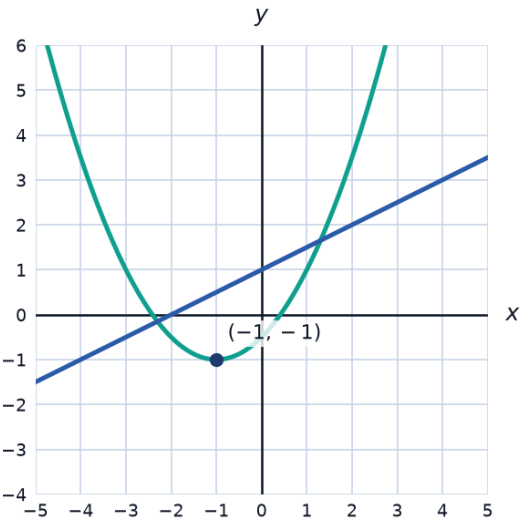
15 [Challenging] • Target: The Work-Energy Principle • Cross-Lesson: 10.01 Series & Binomial Theorem

A body of mass 7 kg moving at 11 m/s has work 470 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

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16 [Challenging] • Target: Power and Vehicles in Motion • Cross-Lesson: 11.01 Kinematics

A body of mass 22 kg moving at 9 m/s has work 159 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.



17 [Challenging] • Target: Work Done by Constant & Inclined Forces • Cross-Lesson: 12.01 Forces & Equilibrium

A body of mass 25 kg moving at 7 m/s has work 448 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

18 [Challenging] • Target: Kinetic Energy & Potential Energy • Cross-Lesson: 13.01 Newton's Laws

A body of mass 8 kg moving at 11 m/s has work 311 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

19 [Challenging] • Target: The Work-Energy Principle • Cross-Lesson: 15.01 Representation of Data

A body of mass 8 kg moving at 8 m/s has work 195 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

20 [Challenging] • Target: Power and Vehicles in Motion • Cross-Lesson: 16.01 Permutations & Combinations

A body of mass 10 kg moving at 7 m/s has work 168 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Work Done by Constant & Inclined Forces

A force of 60 N pulls a box across a horizontal floor through a distance of 8 m. The force acts at an angle of 60° above the horizontal. Find the work done by the force.

a. 240 J

b. 480 J

c. 415.7 J

d. 120 J

22 [Foundational] • Target: Work Done by Constant & Inclined Forces

A crate of mass 15 kg is lifted vertically at a constant speed through a height of 4 m. Taking $g = 10\ \text{m/s}^2$, what is the work done against gravity?

a. 600 J

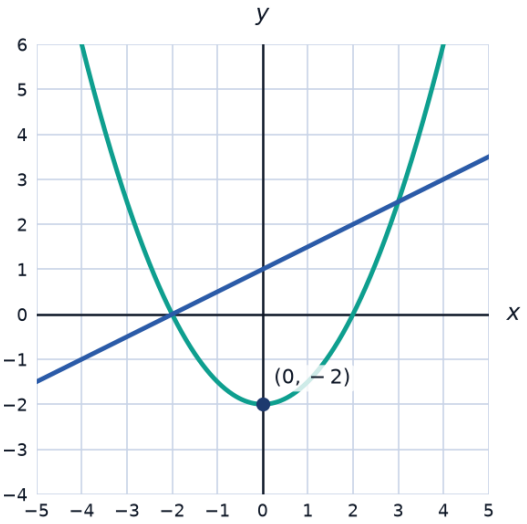
b. 150 J

c. 300 J

d. 60 J

23 [Foundational] • Target: Work Done by Constant & Inclined Forces

A box is pushed 5 m along a horizontal floor against a constant frictional resistance of 35 N. What is the work done against friction?



a. 175 J

b. 150 J

c. 350 J

d. 70 J

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24 [Foundational] • Target: Kinetic Energy & Potential Energy

Calculate the kinetic energy of an athlete of mass 70 kg running at a speed of 8 m/s.

a. 2240 J

b. 4480 J

c. 560 J

d. 280 J

25 [Foundational] • Target: Kinetic Energy & Potential Energy

A rollercoaster car of mass 500 kg starts from rest at the top of a frictionless hill of height 20 m. Taking $g = 10\ \text{m/s}^2$, find its speed at the bottom of the hill.

a. 20 m/s

b. 10 m/s

c. 40 m/s

d. 14.1 m/s

26 [Foundational] • Target: Kinetic Energy & Potential Energy

A bullet of mass 0.02 kg traveling at 300 m/s penetrates a wooden block to a depth of 0.15 m before stopping. What is the average resistive force exerted by the wood?

a. 6000 N

b. 3000 N

c. 9000 N

d. 12000 N

27 [Foundational] • Target: The Work-Energy Principle

A skier of mass 60 kg starts from rest at the top of a slope 30 m high. When reaching the bottom, her speed is 20 m/s. Taking $g = 10\ \text{m/s}^2$, how much energy was lost due to friction?

a. 6000 J

b. 12000 J

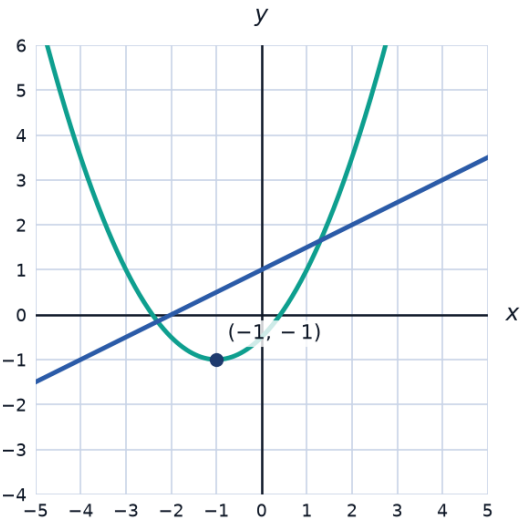
c. 18000 J

d. 3000 J

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28 [Foundational] • Target: The Work-Energy Principle

A package of mass 4 kg slides down a rough incline of length 10 m and vertical drop 6 m. If the resistive force is a constant 8 N, find the speed of the package at the bottom (starting from rest, $g = 10$ m/s^2).



a. $\sqrt{80} \approx 8.94$ m/s

b. 10 m/s

c. 8 m/s

d. 12 m/s

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: The Work-Energy Principle

A particle is projected up a rough incline (30°) with initial speed 10 m/s. If $\mu = 0.2$ and $g = 10$ m/s^2, how far up the plane does the particle travel before coming to rest?

a. ≈ 7.43 m

b. 10.0 m

c. 5.50 m

d. 8.25 m

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30 [Intermediate] • Target: Power and Vehicles in Motion

A car of mass 1200 kg travels along a level road at a constant speed of 30 m/s. The engine is working at 45 kW. What is the total resistance to motion?

a. 1500 N

b. 1200 N

c. 1350 N

d. 2000 N

31 [Intermediate] • Target: Power and Vehicles in Motion

The same car (mass 1200 kg, engine working at 45 kW) faces the same resistance of 1500 N while traveling at 20 m/s. What is its acceleration at this instant?

a. $0.625\ \text{m/s}^2$

b. $0.75\ \text{m/s}^2$

c. $1.00\ \text{m/s}^2$

d. $0.50\ \text{m/s}^2$

32 [Intermediate] • Target: Power and Vehicles in Motion

A truck of mass 5000 kg ascends an incline of 1 in 20 ($\sin\alpha = 1/20$) at a steady speed of 15 m/s. The engine power is 60 kW. Taking $g = 10\ \text{m/s}^2$, find the resistance to motion.

a. 1500 N

b. 2500 N

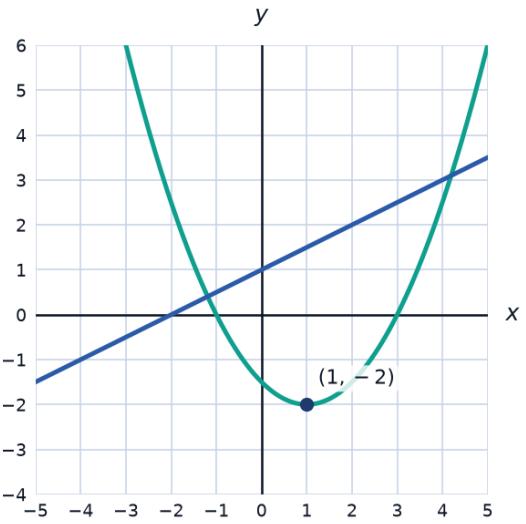
c. 4000 N

d. 2000 N

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33 **[Intermediate]** • **Target: Work Done by Constant & Inclined Forces**

A body of mass 10 kg moving at 4 m/s has work 124 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.



a. 5.39

b. 6.39

c. 9.58

d. 7.39

34 **[Intermediate]** • **Target: Kinetic Energy & Potential Energy**

A body of mass 3 kg moving at 7 m/s has work 218 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

a. 13.94

b. 14.94

c. 20.91

d. 12.94

35 **[Intermediate]** • **Target: The Work-Energy Principle**

A body of mass 7 kg moving at 5 m/s has work 316 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

a. 10.74

b. 16.11

c. 11.74

d. 9.74

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36 [Intermediate] • Target: Power and Vehicles in Motion

A body of mass 15 kg moving at 7 m/s has work 305 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

a. 14.20

b. 10.47

c. 8.47

d. 9.47

37 [Intermediate] • Target: Work Done by Constant & Inclined Forces

A body of mass 24 kg moving at 0 m/s has work 544 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

a. 5.73

b. 6.73

c. 10.10

d. 7.73

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Kinetic Energy & Potential Energy • Cross-Lesson: 13.01 Newton's Laws

A body of mass 6 kg moving at 0 m/s has work 292 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

a. 9.87

b. 14.80

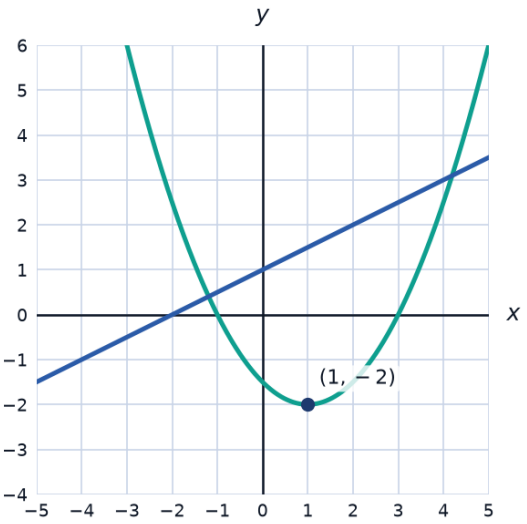
c. 10.87

d. 8.87

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39 [Challenging] • Target: The Work-Energy Principle • Cross-Lesson: 15.01 Representation of Data

A body of mass 7 kg moving at 0 m/s has work 134 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.



a. 6.19

b. 7.19

c. 5.19

d. 9.28

40 [Challenging] • Target: Power and Vehicles in Motion • Cross-Lesson: 16.01 Permutations & Combinations

A body of mass 7 kg moving at 8 m/s has work 45 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

a. 9.77

b. 8.77

c. 13.15

d. 7.77

41 [Challenging] • Target: Work Done by Constant & Inclined Forces • Cross-Lesson: 17.01 Discrete Random Variables

A body of mass 6 kg moving at 8 m/s has work 186 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

a. 11.22

b. 12.22

c. 10.22

d. 16.84

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42 [Challenging] • Target: Kinetic Energy & Potential Energy • Cross-Lesson: 18.01 Binomial Distribution

A body of mass 3 kg moving at 9 m/s has work 282 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

a. 16.40

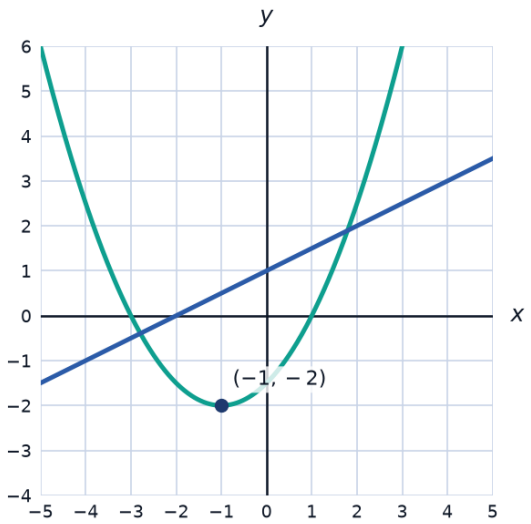
b. 24.60

c. 17.40

d. 15.40

43 [Challenging] • Target: The Work-Energy Principle • Cross-Lesson: 19.01 Normal Distribution

A body of mass 6 kg moving at 7 m/s has work 76 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.



a. 9.62

b. 7.62

c. 8.62

d. 12.93

44 [Challenging] • Target: Power and Vehicles in Motion • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

A body of mass 2 kg moving at 7 m/s has work 99 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

a. 18.25

b. 12.17

c. 11.17

d. 13.17

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45 [Challenging] • Target: Work Done by Constant & Inclined Forces • Cross-Lesson: 3.01 Graphs & Transformations

A body of mass 10 kg moving at 7 m/s has work 200 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

a. 10.43

b. 14.15

c. 8.43

d. 9.43

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